Asymptotic Reversion of $x + \mathrm{W}(x)$

A General Calculus for Logarithmic Transseries

Vladimir Reshetnikov

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Abstract

Let $\mathrm{W} = \mathrm{W}_0$ be the principal real Lambert function and let $g = f^{-1}$, where $f(x) = x + \mathrm{W}(x)$ on $[0, \infty)$. This article derives an all-orders asymptotic expansion of $g(z)$ as $z \to +\infty$, and uses the example to develop a general method for reversing logarithmic transseries near the identity.

The exact substitution $u = \mathrm{W}(g(z))$ reduces the inverse problem to $u\mathrm{e}^u + u = z$. Thus u is the r-Lambert function at $r = 1$, and $g(z) = z - u$. If $v = \mathrm{W}(z)$, then the useful asymptotic expansion is not merely a series in inverse powers of $\log z$, but the finer transseries, which converges for all sufficiently large z,

$$g(z) = z - v + \sum_{m \geq 1} \frac{A_m(v)}{z^m},$$

where every A_m is an explicit rational function. We obtain the operator formula

$$A_m(v) = \frac{(-1)^{m+1}}{(m+1)!} \prod_{j=0}^{m-1} \left(\frac{v}{1+v} \frac{\mathrm{d}}{\mathrm{d}v} - j \right) v^{m+1},$$

a triangular recurrence, a direct Lagrange coefficient formula, and a uniform remainder estimate of order $(\mathrm{W}(z)/z)^{N+1}$ after N correction blocks. A second reversion, in the parameters $q = \mathrm{e}^{-v} = v/z$ and $s = 1/v$, yields harmonic-number formulas for the large-v structure of the coefficient blocks.

At the coarser logarithmic level, $g(z)$ and $z - \mathrm{W}(z)$ have identical complete expansions in powers of $1/\log z$; their difference is beyond all orders of that scale. We give the resulting Stirling-number formula and show precisely how the exponentially small correction must be restored. Finally, for a general near-identity logarithmic transseries $F(X) = X + A(X)$, we prove the locally finite reversion identity

$$F^{-1}(X) = X + \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{\mathrm{d}^{n-1}}{\mathrm{d}X^{n-1}} A(X)^n,$$

derive an explicit block formula over an arbitrary differential coefficient algebra, compare fixed-point, triangular, Lagrange–Bürmann, and Newton algorithms, and give residual certificates that turn formal truncations into rigorous analytic approximations.

Keywords. Lambert W; generalized Lambert function; r-Lambert function; logarithmic transseries; compositional inverse; Lagrange–Bürmann inversion; series reversion; asymptotic scale; harmonic numbers.

Contents

1 The inverse problem and the answer in four forms

Throughout, $\mathrm{W}(x) = \mathrm{W}_0(x)$ is the principal real branch, characterized on $[0, \infty)$ by

$$\mathrm{W}(x)\mathrm{e}^{\mathrm{W}(x)} = x, \qquad \mathrm{W}(x) \geq 0. \tag{1}$$

Define

$$f(x) = x + \mathrm{W}(x), \qquad x \geq 0, \tag{2}$$

and write $g = f^{-1}$.

The standard differentiation formula

$$\mathrm{W}'(x) = \frac{\mathrm{W}(x)}{x(1 + \mathrm{W}(x))} = \frac{\mathrm{e}^{-\mathrm{W}(x)}}{1 + \mathrm{W}(x)}, \qquad x > 0, \tag{3}$$

with $\mathrm{W}'(0) = 1$, shows that $f'(x) > 0$. Since $f(0) = 0$ and $f(x) \to \infty$, the inverse exists uniquely on $[0, \infty)$.

Main result

Put

$$v = \mathrm{W}(z), \qquad L = \log z, \qquad \ell = \log L, \qquad q = \mathrm{e}^{-v} = \frac{v}{z}.$$

Then the inverse $g = f^{-1}$ admits the following complementary descriptions.

1. *Exact generalized-Lambert form:*

$$g(z) = z - \mathcal{W}_1(z), \qquad \mathcal{W}_1(z)\mathrm{e}^{\mathcal{W}_1(z)} + \mathcal{W}_1(z) = z.$$

   Here $\mathcal{W}_r$ denotes the r-Lambert function, not the complex branch W_1 of the ordinary Lambert function.

2. *Derivative or Lagrange–Bürmann form:*

$$g(z) \sim z + \sum_{n \geq 1} \frac{(-1)^n}{n!} \left(\frac{\mathrm{d}}{\mathrm{d}z}\right)^{n-1} \mathrm{W}(z)^n. \tag{4}$$

   For this particular problem the formal expansion reorganizes into a genuinely convergent local inverse series for all sufficiently large positive z.

3. W*-anchored outer expansion:*

$$\begin{aligned} g(z) = z - v &+ \frac{v^2}{z(1+v)} + \frac{v^3(v^2 - 2)}{2z^2(1+v)^3} \\ &+ \frac{v^4(2v-1)(v^3 - 6v - 6)}{6z^3(1+v)^5} + O\left(\left(\frac{v}{z}\right)^4\right). \end{aligned} \tag{5}$$

   An all-orders coefficient formula and six explicit blocks are given below.

4. *Pure logarithmic expansion:*

$$g(z) \sim z - L + \ell - \sum_{n \geq 1} \frac{P_n(\ell)}{L^n}, \tag{6}$$

   where

$$P_n(y) = \sum_{k=1}^{n} (-1)^{n-k} \left[{n \atop n-k+1}\right] \frac{y^k}{k!} \tag{7}$$

   and $\left[{n \atop k}\right]$ is an unsigned Stirling cycle number. This last expansion is complete in the scale $1/L$, but it omits the beyond-all-orders difference between $g(z)$ and $z - \mathrm{W}(z)$.

The rest of the article proves these assertions, explains how they fit into a single transseries, and extracts a reusable reversion calculus.

2 Exact reduction to the r-Lambert function

2.1 The generalized Lambert function

For a fixed real parameter r, the r-Lambert function is locally defined as the inverse of

$$h_r(u) = u\mathrm{e}^u + ru. \tag{8}$$

Following Mező and Baricz [2], we denote it here by $\mathcal{W}_r$. This calligraphic notation prevents confusion with the complex branch symbol W_k for the ordinary Lambert function. When $r \geq 0$, $h_r'(u) = \mathrm{e}^u(1+u) + r > 0$ on $[0, \infty)$, so $\mathcal{W}_r$ is a single-valued increasing function there.

Theorem 2.1 (Exact inverse)**.** *For $z \geq 0$, let $u = \mathcal{W}_1(z)$, the unique nonnegative solution of*

$$u\mathrm{e}^u + u = z. \tag{9}$$

Then

$$\boxed{g(z) = z - u = u\mathrm{e}^u.} \tag{10}$$

Proof. Set $x = g(z)$ and $u = \mathrm{W}(x)$. By (1), $x = u\mathrm{e}^u$. The defining equation $z = f(x) = x + \mathrm{W}(x)$ therefore becomes

$$z = u\mathrm{e}^u + u.$$

Monotonicity of h_1 gives $u = \mathcal{W}_1(z)$. Finally, $x = u\mathrm{e}^u = z - u$, which proves both identities. □

Lean status. Formal, in `LambertShiftInverse.lean`: with `Fabius.lambertShift` the map $f(x) = x + \mathrm{W}(x)$, `Fabius.shiftedLambertW` the root $u \geq 0$ of $u\mathrm{e}^u + u = z$ (defined as the inverse on $[0, \infty)$ of `Fabius.shiftedMulExp`, whose existence is the intermediate value theorem on $[0, z]$ and whose uniqueness is strict monotonicity, `Fabius.shiftedLambertW_unique`), and `Fabius.lambertShiftInv` the difference $z - u$, the theorem is the pair `Fabius.lambertShift_lambertShiftInv` ($f(z-u) = z$) and `Fabius.lambertShiftInv_eq_mul_exp` ($z - u = u\mathrm{e}^u$); the intermediate fact $\mathrm{W}(g(z)) = u$ is `Fabius.principalLambertW_lambertShiftInv`. That g is the two-sided inverse of f on $[0, \infty)$ is `Fabius.lambertShiftInv_lambertShift` together with `Fabius.lambertShift_strictMonoOn` and `Fabius.lambertShift_surjOn`.

Corollary 2.2 (Elementary bounds)**.** *For every $z > 0$,*

$$z - \mathrm{W}(z) < g(z) < z. \tag{11}$$

Proof. The right inequality follows from $g(z) = z - u$ and $u > 0$. Let $v = \mathrm{W}(z)$. Since $v\mathrm{e}^v = z$, one has $h_1(v) = v\mathrm{e}^v + v = z + v > z = h_1(u)$. Monotonicity gives $u < v$, hence $g(z) = z - u > z - v$. □

Lean status. Formal: `Fabius.lambertShiftInv_lt_self` and `Fabius.sub_principalLambertW_lt_lambertShiftInv` in `LambertShiftInverse.lean`, for every $z > 0$; the second is the comparison $u < \mathrm{W}(z)$ (`Fabius.shiftedLambertW_lt_principalLambertW`) read off the strict monotonicity of $t \mapsto t\mathrm{e}^t$.

Differentiating (9) gives

$$u'(z) = \frac{1}{\mathrm{e}^u(1+u)+1}, \qquad g'(z) = \frac{\mathrm{e}^u(1+u)}{\mathrm{e}^u(1+u)+1} = \frac{1+u}{1+u+\mathrm{e}^{-u}}. \tag{12}$$

Thus $0 < g'(z) < 1$, as expected from a strictly increasing perturbation of the identity.

2.2 The comparison point $v = \mathrm{W}(z)$

The ordinary Lambert value $v = \mathrm{W}(z)$ solves

$$v\mathrm{e}^v = z. \tag{13}$$

Since $u < v$, define the positive displacement

$$\tau = v - u. \tag{14}$$

The inverse can then be written as

$$g(z) = z - v + \tau. \tag{15}$$

The entire problem is reduced to finding the small quantity τ. Substituting $u = v - \tau$ in (9) and dividing by e^v gives

$$(v - \tau)\mathrm{e}^{-\tau} + (v - \tau)\mathrm{e}^{-v} = v. \tag{16}$$

With

$$q = \mathrm{e}^{-v} = \frac{v}{z}, \tag{17}$$

this becomes the exact inverse-series equation

$$q = \Phi_v(\tau), \qquad \Phi_v(\tau) = \frac{v}{v - \tau} - \mathrm{e}^{-\tau}. \tag{18}$$

Since $\Phi_v(0) = 0$ and

$$\Phi_v'(0) = 1 + \frac{1}{v} > 0, \tag{19}$$

Φ_v has a local analytic inverse at the origin. Equation (18) will later provide both convergence and efficient coefficient generation.

3 Why two asymptotic scales are necessary

Let

$$L = \log z, \qquad \ell = \log L. \tag{20}$$

The ordinary Lambert function has the familiar size $v = L - \ell + o(1)$. Consequently

$$q = \frac{v}{z} \sim \frac{L}{z}. \tag{21}$$

For every fixed N,

$$\frac{L}{z} = o(L^{-N}). \tag{22}$$

Thus the displacement τ, which begins at order q, is smaller than every power of $1/L$.

This yields a strict hierarchy:

$$z \gg L \gg \ell \gg 1 \gg \frac{\ell}{L} \gg \frac{\ell^2}{L^2} \gg \cdots \gg \frac{L}{z} \gg \frac{L^2}{z^2} \gg \cdots.$$

The first line of asymptotics, organized only by powers of $1/L$, determines $z - v$. It cannot detect τ. The second line, organized by powers of $q = v/z$, resolves the difference between $z - v$ and the true inverse.

Scale warning

A statement such as

$$g(z) = z - L + \ell - \frac{\ell}{L} + O\left(\frac{\ell^2}{L^2}\right)$$

is correct but incapable of revealing the correction $\tau \asymp L/z$, because that correction is already swallowed by every finite inverse-logarithm remainder. A complete answer must therefore either retain $v = \mathrm{W}(z)$ as an exact slow variable or use a nested transseries with an outer z^{-1} scale and inner logarithmic coefficient expansions.

4 A differential algebra for logarithmic transseries

The special function calculation is an instance of a general formal mechanism. We develop that mechanism before returning to W.

4.1 Slow coefficient algebras and the outer variable

Let $X \to +\infty$, put $t = X^{-1}$, and introduce iterated logarithms

$$L_1 = \log X, \qquad L_2 = \log L_1, \qquad L_3 = \log L_2, \ \ldots. \tag{23}$$

Let $\mathcal{C}$ be a commutative $\mathbb{Q}$-algebra of slow coefficients containing whatever rational functions, logarithmic polynomials, or completed inner series are required. It is equipped with the derivation

$$\delta = \frac{\partial}{\partial L_1} + \frac{1}{L_1}\frac{\partial}{\partial L_2} + \frac{1}{L_1 L_2}\frac{\partial}{\partial L_3} + \cdots, \tag{24}$$

truncated after the last iterated logarithm actually present. Then $\mathrm{d}c/\,\mathrm{d}X = t\,\delta c$ for $c \in \mathcal{C}$.

Consider the outer formal series ring

$$\mathcal{C}[[t]] = \left\{ \sum_{n\ge 0} c_n t^n : c_n \in \mathcal{C} \right\}. \tag{25}$$

The derivative with respect to X acts as

$$D := \frac{\mathrm{d}}{\mathrm{d}X} = t\delta - t^2\frac{\partial}{\partial t}. \tag{26}$$

In particular,

$$D(ct^r) = t^{r+1}(\delta - r)c. \tag{27}$$

Every differentiation raises the outer t-valuation by one. This simple fact is responsible for all local finiteness below.

Lemma 4.1 (Repeated differentiation). *For $c \in \mathcal{C}$, $r \ge 0$, and $k \ge 0$,*

$$D^k(ct^r) = t^{r+k}\prod_{j=0}^{k-1}(\delta - r - j)c, \tag{28}$$

where the empty product for $k = 0$ is the identity.

Proof. The case $k = 0$ is immediate, and (27) is the case $k = 1$. If the formula holds for k, apply (27) to the coefficient of t^{r+k}. This appends the commuting factor $\delta - r - k$. □

4.2 Composition near the identity

For $A, B \in \mathcal{C}[[t]]$, composition of $A(X)$ with $X + B(X)$ is defined by the formal Taylor series

$$A\big(X + B(X)\big) := \sum_{k\geq 0} \frac{B(X)^k}{k!} D^k A(X). \tag{29}$$

Although B may have outer order zero, $D^k A$ has order at least k. Hence only finitely many k contribute to each coefficient of t^N, so (29) is well-defined. This is the algebraic version of expanding

$$\log(X + B) = \log X + \log(1 + tB)$$

and then expanding all subsequent iterated logarithms.

4.3 Lagrange–Bürmann reversion at infinity

Theorem 4.2 (Near-identity reversion)**.** *Let*

$$F(X) = X + A(X), \qquad A \in \mathcal{C}[[t]]. \tag{30}$$

Then F has a unique compositional inverse of the form $G(X) = X + B(X)$, $B \in \mathcal{C}[[t]]$, and

$$\boxed{G(X) = X + \sum_{n\geq 1} \frac{(-1)^n}{n!} D^{n-1}\big(A(X)^n\big).} \tag{31}$$

The series is locally finite in the outer t-adic topology: its n-th summand has valuation at least $n-1$.

Proof. Introduce a formal parameter ε and solve

$$G_\varepsilon(X) + \varepsilon A(G_\varepsilon(X)) = X.$$

Writing $G_\varepsilon = X + Y$, this is $Y = -\varepsilon A(X + Y)$. The classical Lagrange–Bürmann formula, applied as a formal power series in ε, gives

$$Y = \sum_{n\geq 1} \frac{(-\varepsilon)^n}{n!} D^{n-1} A(X)^n.$$

By (26), the coefficient of ε^n has outer valuation at least $n-1$. Therefore setting $\varepsilon = 1$ is legitimate: for any prescribed outer order, only finitely many values of n occur. The resulting series satisfies both composition identities by formal substitution.

For uniqueness, suppose two inverses differ first at order t^N. Since $F'(X) = 1 + O(t)$, formal Taylor expansion shows that their images under F differ by the same nonzero leading t^N block, a contradiction. ☐

The first universal terms of (31) are

$$\begin{aligned} G = X - A + AA' - A(A')^2 - \frac{A^2}{2}A'' \\ + A(A')^3 + \frac{3}{2}A^2A'A'' + \frac{A^3}{6}A''' + \cdots, \end{aligned} \tag{32}$$

where primes mean D. This display is useful for hand calculations, but the compact formula (31) is safer at high order.

Remark 4.3 (Bell-polynomial form). Let $B_{m,k}$ be the partial exponential Bell polynomial. For $n \geq 2$,

$$D^{n-1}(A^n) = \sum_{k=1}^{n-1} (n)_{\underline{k}} A^{n-k} B_{n-1,k}(A', A'', \ldots, A^{(n-k)}), \tag{33}$$

which is a direct Faà di Bruno expansion of $D^{n-1}(y^n)|_{y=A}$. Thus every universal reversion coefficient can be generated by standard Bell polynomial machinery.

4.4 An explicit coefficient-block formula

Write

$$A = \sum_{r\geq 0} a_r t^r, \qquad A^n = \sum_{r\geq 0} C_{n,r} t^r, \qquad C_{n,r} = \sum_{r_1+\cdots+r_n=r} a_{r_1}\cdots a_{r_n}. \tag{34}$$

If $G = X + \sum_{N\geq 0} b_N t^N$, then Theorems 4.1 and 4.2 give the finite formula

$$\boxed{b_N = \sum_{n=1}^{N+1} \frac{(-1)^n}{n!} \left[\prod_{j=0}^{n-2} (\delta - (N-n+1) - j)\right] C_{n,N-n+1}.} \tag{35}$$

The empty product at $n = 1$ is one. The first blocks are

$$b_0 = -a_0, \tag{36}$$

$$b_1 = -a_1 + a_0 \delta a_0, \tag{37}$$

$$b_2 = -a_2 + (\delta - 1)(a_0 a_1) - \frac{1}{6}\delta(\delta-1)(a_0^3), \tag{38}$$

$$\begin{aligned} b_3 = &-a_3 + \frac{1}{2}(\delta-2)(2a_0a_2 + a_1^2) \\ &- \frac{1}{6}(\delta-1)(\delta-2)(3a_0^2 a_1) + \frac{1}{24}\delta(\delta-1)(\delta-2)(a_0^4). \end{aligned} \tag{39}$$

Reusable method

To reverse any logarithmic transseries $F = X + A$: choose a slow differential coefficient algebra $(\mathcal{C}, \delta)$, expand $A = \sum a_r t^r$, compute the finite convolutions $C_{n,r}$, and apply (35). No guessing of logarithmic ansatzes is required, and each output block depends on only finitely many input blocks.

5 Application of the general formula to $A(X) = \mathrm{W}(X)$

The exact slow variable

$$v = \mathrm{W}(X) \tag{40}$$

forms a particularly economical differential coefficient field. Since $X = ve^v$,

$$\frac{\mathrm{d}v}{\mathrm{d}X} = \frac{v}{X(1+v)} = t\,\frac{v}{1+v}. \tag{41}$$

Thus, on rational functions of v, the slow derivation is

$$\vartheta := \frac{v}{1+v}\frac{\mathrm{d}}{\mathrm{d}v}, \qquad D = t\vartheta - t^2\frac{\partial}{\partial t}. \tag{42}$$

In this coefficient field $A = v$ has only one outer block, namely $a_0 = v$ and $a_r = 0$ for $r \geq 1$. Therefore, in the block formula (35), exactly one Lagrange term contributes to each outer order.

Theorem 5.1 (All-orders W-anchored expansion). *Let $v = \mathrm{W}(z)$. Then, formally in z^{-1},*

$$g(z) = z - v + \sum_{m\geq 1} \frac{A_m(v)}{z^m}, \tag{43}$$

where

$$\boxed{A_m(v) = \frac{(-1)^{m+1}}{(m+1)!} \prod_{j=0}^{m-1} (\vartheta - j) v^{m+1}, \qquad \vartheta = \frac{v}{1+v}\frac{\mathrm{d}}{\mathrm{d}v}.} \tag{44}$$

Equivalently,

$$g(z) = z + \sum_{n\geq 1} \frac{(-1)^n}{n!} \left(\frac{\mathrm{d}}{\mathrm{d}z}\right)^{n-1} v^n. \tag{45}$$

Proof. Apply (31) with $A = v$. For a rational function $R(v)$,

$$\frac{\mathrm{d}}{\mathrm{d}z}\left(z^{-j}R(v)\right) = z^{-j-1}(\vartheta - j)R(v). \tag{46}$$

Repeated application to $R(v) = v^{m+1}$ proves (44) after putting $n = m+1$. The $n = 1$ term is $-v$. ☐

The first six coefficient functions are as follows:

$$A_1(v) = \frac{v^2}{1+v}, \tag{47}$$

$$A_2(v) = \frac{v^3(v^2-2)}{2(1+v)^3}, \tag{48}$$

$$A_3(v) = \frac{v^4(2v-1)(v^3-6v-6)}{6(1+v)^5}, \tag{49}$$

$$A_4(v) = \frac{v^5}{24(1+v)^7}\left(6v^6 - 8v^5 - 69v^4 - 40v^3 + 96v^2 + 72v - 24\right), \tag{50}$$

$$\begin{aligned} A_5(v) = \frac{v^6}{120(1+v)^9}\big(&24v^8 - 58v^7 - 428v^6 - 151v^5 + 1320v^4 \\ &+ 1380v^3 - 480v^2 - 720v + 120\big), \end{aligned} \tag{51}$$

$$\begin{aligned} A_6(v) = \frac{v^7}{720(1+v)^{11}}\big(&120v^{10} - 444v^9 - 2920v^8 + 424v^7 + 16577v^6 \\ &+ 17892v^5 - 13530v^4 - 26400v^3 - 1800v^2 + 7200v - 720\big). \end{aligned} \tag{52}$$

Substitution in (43) gives a practical high-accuracy expansion without expanding v into logarithms.

5.1 Derivative polynomials and a triangular generator

For symbolic computation it is useful to avoid repeated rational differentiation from scratch. Define polynomials $R_{n,k}(v)$ by

$$\left(\frac{\mathrm{d}}{\mathrm{d}z}\right)^k v^n = \frac{1}{z^k}\frac{v^n R_{n,k}(v)}{(1+v)^{2k-1}}, \qquad k \geq 1. \tag{53}$$

Then

$$R_{n,1}(v) = n, \tag{54}$$

and differentiation of (53) gives

$$\begin{aligned} R_{n,k+1}(v) &= v(1+v)R'_{n,k}(v) \\ &\quad + \left(n(1+v) - (2k-1)v - k(1+v)^2\right)R_{n,k}(v). \end{aligned} \tag{55}$$

Consequently,

$$A_m(v) = \frac{(-1)^{m+1}}{(m+1)!}\frac{v^{m+1}R_{m+1,m}(v)}{(1+v)^{2m-1}}. \tag{56}$$

This two-index recurrence generates the diagonal needed for the inverse and, at the same time, all derivatives of all powers of W.

Proposition 5.2 (Integer-polynomial structure)**.** *For every $m \geq 1$, there is a polynomial $P_m \in \mathbb{Z}[v]$ such that*

$$A_m(v) = \frac{v^{m+1} P_m(v)}{m!(1+v)^{2m-1}}. \tag{57}$$

Moreover,

$$\deg P_m = 2m - 2, \qquad [v^{2m-2}]P_m = (m-1)!, \qquad P_m(0) = (-1)^{m+1} m!. \tag{58}$$

In particular,

$$A_m(v) \sim \frac{v^m}{m} \qquad (v \to +\infty). \tag{59}$$

Proof. Fix a positive integer n. Since $R_{n,1} = n$ and every operation on the right-hand side of (55) preserves integer coefficients and divisibility by n, one has $R_{n,k} \in n\mathbb{Z}[v]$ for every $k \geq 1$. Consequently

$$P_m(v) = \frac{(-1)^{m+1}}{m+1} R_{m+1,m}(v)$$

is an integer polynomial, and (57) follows from (56).

It remains to determine its extremal coefficients. If $d_k = \deg R_{n,k}$ and r_k is the leading coefficient, then (55) shows, inductively, that

$$d_1 = 0, \qquad d_{k+1} = d_k + 2, \qquad r_{k+1} = -k r_k.$$

Indeed, the unique term of degree $d_k + 2$ is $-kv^2 R_{n,k}$. Hence

$$\deg R_{n,k} = 2k - 2, \qquad [v^{2k-2}]R_{n,k} = (-1)^{k-1}(k-1)!n.$$

On the diagonal $n = m+1$, $k = m$, multiplication by $(-1)^{m+1}/(m+1)$ gives $\deg P_m = 2m - 2$ and leading coefficient $(m-1)!$.

Finally, evaluating (55) at $v = 0$ gives

$$R_{n,k+1}(0) = (n-k)R_{n,k}(0), \qquad R_{n,1}(0) = n.$$

Therefore $R_{n,k}(0) = n!/(n-k)!$ for $1 \leq k \leq n$, and

$$P_m(0) = \frac{(-1)^{m+1}}{m+1}(m+1)! = (-1)^{m+1} m!.$$

Substitution in (57) now gives $A_m(v) \sim v^m/m$, completing the proof. □

Lean status. The polynomial family and every item of (58) are formal in `LambertInverseCoefficients.lean`, as algebra over $\mathbb{Z}[v]$. With `Fabius.lambertNumerator` the family $R_{n,k}$ (indices shifted so the recursion starts at 0: `Fabius.lambertNumerator` $n\,j$ is $R_{n,j+1}$), the initial value (54) is the definition and the recurrence (55) is `Fabius.lambertNumerator_succ_eq_draft`, written exactly as displayed. From it: `Fabius.natCast_dvd_lambertNumerator` is $R_{n,k} \in n\mathbb{Z}[v]$, which is what makes $P_m = (-1)^{m+1} R_{m+1,m}/(m+1)$ an integer polynomial; `Fabius.natDegree_lambertNumerator` is $\deg R_{n,k} = 2(k-1)$, giving $\deg P_m = 2m-2$; `Fabius.leadingCoeff_lambertNumerator` is $[v^{2(k-1)}]R_{n,k} = n(-1)^{k-1}(k-1)!$, which at $n = m+1$, $k = m$ gives $[v^{2m-2}]P_m = (m-1)!$; and `Fabius.eval_zero_lambertNumerator` is $R_{n,k}(0) = n(n-1)\cdots(n-k+1)$, which at $n = m+1$, $k = m$ is $(m+1)!$, giving $P_m(0) = (-1)^{m+1} m!$. What is not formalized is the analytic half: the closed form (53) for $(\mathrm{d}/\mathrm{d}z)^k v^n$, hence the identification (56) of this family with A_m, and the asymptotic reading (59). Those need the operator calculus on functions, not the recursion.

6 Direct reversion in the exponentially small parameter

The formal Lagrange expansion already gives the answer. The exact equation (18) gives more: a convergent series for all sufficiently large z, a second coefficient recurrence, and a natural remainder scale.

6.1 Lagrange inversion of Φ_v

Let

$$\tau(q; v) = \sum_{n \geq 1} c_n(v) q^n \tag{60}$$

be the local inverse of $q = \Phi_v(\tau)$. Lagrange inversion gives

$$\boxed{c_n(v) = \frac{1}{n}[\tau^{n-1}]\left(\frac{\tau}{\Phi_v(\tau)}\right)^n = \frac{1}{n!}\,\frac{\mathrm{d}^{n-1}}{\mathrm{d}\tau^{n-1}}\left(\frac{\tau}{\Phi_v(\tau)}\right)^n\Bigg|_{\tau=0}.} \tag{61}$$

Since $q = v/z$, comparison with (43) yields

$$A_n(v) = v^n c_n(v). \tag{62}$$

For example,

$$c_1(v) = \frac{v}{1+v}, \tag{63}$$

$$c_2(v) = \frac{v(v^2-2)}{2(1+v)^3}, \tag{64}$$

$$c_3(v) = \frac{v(2v-1)(v^3-6v-6)}{6(1+v)^5}. \tag{65}$$

Expanding Φ_v itself,

$$\Phi_v(\tau) = \sum_{k \geq 1} \phi_k(v)\tau^k, \qquad \phi_k(v) = \frac{1}{v^k} - \frac{(-1)^k}{k!}, \tag{66}$$

produces a purely triangular recurrence. If $C(q) = \sum_{n \geq 1} c_n q^n$, then the identity $\sum_{k \geq 1} \phi_k C(q)^k = q$ gives

$$c_1 = \frac{1}{\phi_1}, \tag{67}$$

$$c_n = -\frac{1}{\phi_1}\sum_{k=2}^{n} \phi_k \sum_{\substack{j_1+\cdots+j_k=n \\ j_i \geq 1}} c_{j_1}\cdots c_{j_k}, \qquad n \geq 2. \tag{68}$$

All arithmetic is rational in v.

6.2 Convergence and a natural remainder theorem

Theorem 6.1 (Convergent outer expansion for large z)**.** *There exist constants $v_0 > 0$, $\rho > 0$, and $M_N > 0$ such that for all real $v \geq v_0$, the inverse $\tau(q; v) = \Phi_v^{-1}(q)$ is analytic for $|q| < \rho$, with ρ independent of v. Consequently, for $z = v\mathrm{e}^v$ and every fixed N,*

$$g(z) = z - v + \sum_{m=1}^{N} \frac{A_m(v)}{z^m} + O_N\left(\left(\frac{v}{z}\right)^{N+1}\right), \qquad z \to +\infty. \tag{69}$$

For all sufficiently large z, the infinite series (43) converges to $g(z)$.

Proof. Put $s = 1/v$ and write

$$\Phi_s(\tau) = \frac{1}{1 - s\tau} - e^{-\tau}. \tag{70}$$

This is analytic in (s, τ) near $(0, 0)$, and

$$\Phi_s(0) = 0, \qquad \partial_\tau \Phi_s(0) = 1 + s.$$

The analytic inverse function theorem with parameter therefore gives a bidisc $|s| < s_0$, $|q| < \rho$ on which the inverse $\tau = T(q, s)$ is analytic. Compactness after slightly shrinking the bidisc makes ρ uniform for $0 \le s \le s_0/2$, equivalently $v \ge v_0$.

Now the actual parameter is $q = e^{-v}$, which is eventually smaller than ρ. Taylor's theorem in q, uniformly for the indicated values of s, gives a remainder bounded by $M_N|q|^{N+1}$. Substituting $q = v/z$, $\tau = g - (z - v)$, and $A_m = v^m c_m$ proves (69). The same analytic inverse theorem proves convergence of the full Taylor series. □

Remark 6.2. The proof is intentionally local. It neither requires nor asserts a simple closed form for the exact radius of convergence as a function of v. For asymptotics, a uniform positive radius is sufficient because the actual argument $q = e^{-v}$ tends to zero exponentially.

6.3 A bivariate re-expansion and harmonic numbers

The parameter form (70) permits another useful organization: expand first in $s = 1/v$, while retaining the entire dependence on q. Write

$$T(q, s) = \sum_{j \ge 0} s^j T_j(q), \qquad E = 1 - q. \tag{71}$$

At $s = 0$, $q = 1 - e^{-T_0}$, so

$$T_0(q) = -\log(1 - q). \tag{72}$$

Coefficient comparison in $1/(1 - sT) - e^{-T} = q$ gives successively

$$T_1(q) = \frac{\log(1 - q)}{1 - q}, \tag{73}$$

$$T_2(q) = \frac{-\log(1 - q) + \frac{1}{2}\log^2(1 - q)}{(1 - q)^2} - \frac{\log^2(1 - q)}{1 - q}. \tag{74}$$

Thus

$$\begin{aligned} \tau = {} & -\log(1 - q) + \frac{1}{v}\frac{\log(1 - q)}{1 - q} \\ & + \frac{1}{v^2}\left(\frac{-\log(1 - q) + \frac{1}{2}\log^2(1 - q)}{(1 - q)^2} - \frac{\log^2(1 - q)}{1 - q}\right) + O(v^{-3}). \end{aligned} \tag{75}$$

This formula resums infinitely many powers of q at each displayed order in $1/v$.

The construction is triangular to every order. Indeed, at order s^j, the unknown T_j occurs only through the linear term $(1 - q)T_j$ generated by $-e^{-T}$. Hence T_j is obtained by dividing an explicit expression in $T_0, \dots, T_{j-1}$ by $1 - q$. Inductively,

$$T_j(q) \in (1 - q)^{-j}\mathbb{Q}[1 - q, \log(1 - q)]. \tag{76}$$

Let

$$H_n = \sum_{k=1}^{n} \frac{1}{k}, \qquad H_n^{(2)} = \sum_{k=1}^{n} \frac{1}{k^2}. \tag{77}$$

The generating identities

$$
\begin{aligned}
-\log(1-q) &= \sum_{n\ge 1} \frac{q^n}{n}, \\
\frac{\log(1-q)}{1-q} &= -\sum_{n\ge 1} H_n q^n, \\
\frac{\log^2(1-q)}{1-q} &= \sum_{n\ge 1} \big(H_n^2 - H_n^{(2)}\big) q^n
\end{aligned}
\tag{78}
$$

then imply the uniform large-v coefficient expansion

$$
\boxed{c_n(v) = \frac{1}{n} - \frac{H_n}{v} + \frac{H_n + \frac{n-1}{2}\big(H_n^2 - H_n^{(2)}\big)}{v^2} + O_n(v^{-3}).}
\tag{79}
$$

The leading limit $c_n(v) \to 1/n$ reflects the limiting inverse $T_0(q) = -\log(1-q)$.

7 The complete inverse-logarithm expansion

The ordinary Lambert function has the following expansion, which converges for all sufficiently large positive z [1, 3]:

$$
v = \mathrm{W}(z) = L - \ell + \sum_{n\ge 1} \frac{P_n(\ell)}{L^n},
\tag{80}
$$

with P_n given by (7). The first polynomials are

$$
\begin{aligned}
P_1(y) &= y, \\
P_2(y) &= \frac{y(y-2)}{2}, \\
P_3(y) &= \frac{y(2y^2 - 9y + 6)}{6}, \\
P_4(y) &= \frac{y(3y^3 - 22y^2 + 36y - 12)}{12}, \\
P_5(y) &= \frac{y(12y^4 - 125y^3 + 350y^2 - 300y + 60)}{60}, \\
P_6(y) &= \frac{y(20y^5 - 274y^4 + 1125y^3 - 1700y^2 + 900y - 120)}{120}.
\end{aligned}
\tag{81}
$$

Theorem 7.1 (Pure logarithmic asymptotics of the inverse)**.** *For every fixed $N \ge 0$,*

$$
g(z) = z - L + \ell - \sum_{n=1}^{N} \frac{P_n(\ell)}{L^n} + O_N\!\left(\frac{\ell^{N+1}}{L^{N+1}}\right), \qquad z \to +\infty.
\tag{82}
$$

Equivalently, $g(z)$ and $z - \mathrm{W}(z)$ have the same complete asymptotic expansion in the inverse-logarithm scale.

Proof. By Theorem 6.1,

$$
g(z) - (z - v) = O(v/z) = O(L/z).
$$

Equation (22) shows that $L/z = o(L^{-M})$ for every fixed M. Therefore the difference is beyond all orders in $1/L$, and substitution of the standard Lambert expansion (80) proves the result. ☐

Lean status. Partial. The Lambert-side input, the leading two terms of (80), is formal in `PrincipalLambertWAtTop.lean`: the identity $\mathrm{W}(x) + \log \mathrm{W}(x) = \log x$ (`Fabius.principalLambertW_add_log`), the bracket $L - \ell \le \mathrm{W}(x) \le L$ for $x \ge \mathrm{e}$ (`Fabius.principalLambertW_log_bracket`), $\mathrm{W} \sim \log$ at $+\infty$ (`Fabius.principalLambertW_isEquivalent_log`) and the two-term statement $\mathrm{W}(x) - (L - \ell) \to 0$ (`Fabius.tendsto_principalLambertW_sub_log_sub_log_log`). The higher terms $P_n(\ell)/L^n$, and the beyond-all-orders step $g(z) - (z - \mathrm{W}(z)) = O(L/z)$ that transfers the expansion from $z - \mathrm{W}(z)$ to g, are not formalized.

In explicit form,

$$
\begin{aligned}
g(z) \sim z - L + \ell - \frac{\ell}{L} - \frac{\ell(\ell-2)}{2L^2} - \frac{\ell(2\ell^2 - 9\ell + 6)}{6L^3} \\
- \frac{\ell(3\ell^3 - 22\ell^2 + 36\ell - 12)}{12L^4} \\
- \frac{\ell(12\ell^4 - 125\ell^3 + 350\ell^2 - 300\ell + 60)}{60L^5} \\
- \frac{\ell(20\ell^5 - 274\ell^4 + 1125\ell^3 - 1700\ell^2 + 900\ell - 120)}{120L^6} + \cdots .
\end{aligned}
\tag{83}
$$

What the complete slow expansion still does not know

Even if the series (80) is summed to the exact value $v = \mathrm{W}(z)$, it gives only $z - v$, not $g(z)$. The missing positive amount is

$$
\tau = g(z) - (z - v) = \frac{v^2}{z(1+v)} + O\left(\frac{v^2}{z^2}\right) \sim \frac{\log z}{z}.
$$

Thus the generalized inverse contains information that is invisible to the entire rank-one inverse-logarithm expansion. This is a concrete example of why asymptotic equality to all algebraic orders does not imply equality of functions.

8 The full two-level logarithmic transseries

The most informative answer is the nested expansion

$$
\boxed{g(z) \sim_{\mathcal{T}} z - v + \sum_{m \ge 1} \frac{A_m(v)}{z^m}, \qquad v \sim L - \ell + \sum_{n \ge 1} \frac{P_n(\ell)}{L^n}.}
\tag{84}
$$

The symbol $\sim_{\mathcal{T}}$ emphasizes a lexicographically ordered transseries: first by the outer exponent of z^{-1}, and inside each outer block by the inverse powers of L, with polynomial dependence on ℓ. Keeping $v = \mathrm{W}(z)$ exact is usually computationally superior; expanding v is useful when only elementary logarithms are desired.

8.1 Leading inner structure of every outer block

Combining (62) and (79) gives

$$
A_m(v) = \frac{v^m}{m} - H_m v^{m-1} + d_m v^{m-2} + O_m(v^{m-3}),
\tag{85}
$$

where

$$
d_m = H_m + \frac{m-1}{2}\left(H_m^2 - H_m^{(2)}\right).
\tag{86}
$$

Using $v = L - \ell + \ell/L + O(\ell^2/L^2)$, one obtains a uniform formula for the first three inner layers of every outer block:

$$A_m(\mathrm{W}(z)) = \frac{L^m}{m} - (\ell + H_m)L^{m-1} \\ + \left[\frac{m-1}{2}\ell^2 + \big(1 + (m-1)H_m\big)\ell + d_m\right] L^{m-2} + O_m(L^{m-3}\ell^3). \tag{87}$$

For the first three blocks this reads

$$A_1(\mathrm{W}(z)) = L - \ell - 1 + \frac{\ell + 1}{L} + O\left(\frac{\ell^2}{L^2}\right), \tag{88}$$

$$A_2(\mathrm{W}(z)) = \frac{L^2}{2} - \left(\ell + \frac{3}{2}\right) L + \frac{\ell^2}{2} + \frac{5}{2}\ell + 2 + O\left(\frac{\ell^2}{L}\right), \tag{89}$$

$$A_3(\mathrm{W}(z)) = \frac{L^3}{3} - \left(\ell + \frac{11}{6}\right) L^2 + \left(\ell^2 + \frac{14}{3}\ell + \frac{23}{6}\right) L + O(\ell^3). \tag{90}$$

Hence the first beyond-all-orders sector begins as

$$\frac{A_1(v)}{z} = \frac{L - \ell - 1}{z} + \frac{\ell + 1}{zL} + O\left(\frac{\ell^2}{zL^2}\right), \tag{91}$$

and the next sector begins with $L^2/(2z^2)$.

Remark 8.1 (Why the leading coefficients are logarithmic). At fixed m, $A_m(v)/z^m \sim q^m/m$. Summing only these leading pieces over all m gives

$$\sum_{m \geq 1} \frac{q^m}{m} = -\log(1 - q),$$

which is exactly the $s = 0$ resummation $T_0(q)$ in (72). The harmonic correction $-H_m/v$ in (79) resums to $T_1(q)/v$. Thus the rational coefficient blocks, the harmonic numbers, and the elementary logarithm $-\log(1 - q)$ are three views of the same bivariate inverse.

8.2 A transseries ordering convention

A practical truncation is specified by two cutoffs. Choose an outer order M and, for each $0 \leq m \leq M$, an inner order N_m. Then:

1. expand the slow block $-v$ through L^{-N_0};
2. for $1 \leq m \leq M$, expand $A_m(v)$ through L^{m-N_m}, then multiply by z^{-m};
3. retain the unexpanded form $A_m(\mathrm{W}(z))/z^m$ whenever numerical evaluation of W is available.

A single scalar "order" is insufficient because an error such as L^{-100} is still much larger than the leading L/z sector. This is not a defect; it is the defining feature of a rank-two asymptotic scale.

9 Algorithms for general logarithmic-transseries reversion

The closed formula (31) is conceptually definitive, but several algorithms are useful in different computational regimes.

9.1 Algorithm I: direct Lagrange–Bürmann blocks

Algorithm 9.1 (Explicit block construction)**.** Given $A = \sum_{r=0}^{N} a_r t^r + O(t^{N+1})$:

1. form $C_{n,r} = [t^r]A^n$ for $1 \le n \le N+1$ and $0 \le r \le N-n+1$;
2. apply the falling differential operator in (35);
3. sum over n to obtain b_N, and repeat for all lower blocks if they have not already been computed.

The output is the inverse $G = X + \sum_{r=0}^{N} b_r t^r + O(t^{N+1})$.

This method gives transparent closed formulas and is ideal when exact coefficient identities matter. Its disadvantage is repeated formation of powers A^n if implemented naively.

9.2 Algorithm II: triangular residual cancellation

Suppose an approximation $G_{<N}$ has been constructed with

$$F(G_{<N}) - X = r_N t^N + O(t^{N+1}). \tag{92}$$

Because $F'(G_{<N}) = 1 + O(t)$, the correction

$$G_{<N+1} = G_{<N} - r_N t^N \tag{93}$$

raises the residual order by at least one. Starting with $G_{<0} = X$, this builds the inverse one block at a time. The first step automatically gives $b_0 = -a_0$.

Algorithm 9.2 (Triangular reversion)**.** For $N = 0, 1, 2, \ldots$: compose F with the current truncation, extract the first nonzero residual block r_N, subtract $r_N t^N$ from the current inverse, and truncate all intermediate expressions above the target order.

The triangular method is often the simplest robust implementation. It also provides a residual certificate at every stage.

9.3 Algorithm III: fixed-point iteration

The inverse equation is

$$G = X - A(G). \tag{94}$$

Starting with $G_0 = X$, define

$$G_{k+1} = X - A(G_k). \tag{95}$$

If two approximants first differ at order t^N, composition with A raises that difference to order at least t^{N+1}, because $A' = O(t)$. Thus each iteration gains at least one outer block. This is a formal contraction in the t-adic metric.

For the present problem,

$$G_{k+1}(z) = z - \mathrm{W}(G_k(z)). \tag{96}$$

Numerically this is already contractive for every positive argument, since $0 < \mathrm{W}'(x) \le 1$, and it becomes rapidly contractive as $x \to \infty$.

9.4 Algorithm IV: Newton reversion

Newton's method applies directly to $F(G) - X = 0$:

$$G_{\mathrm{new}} = G - \frac{F(G) - X}{F'(G)}. \tag{97}$$

In the filtered transseries algebra, reciprocal computation is legitimate because $F'(G) = 1 + O(t)$ is a unit. If the residual has outer order N, the new residual has at least approximately twice that precision; for a slow perturbation, $F'' = O(t^2)$ makes the gain even stronger.

For $F(x) = x + \mathrm{W}(x)$,

$$G_{\text{new}} = G - \frac{G + \mathrm{W}(G) - z}{1 + \mathrm{W}(G)/(G(1 + \mathrm{W}(G)))}. \tag{98}$$

A low-order transseries seed followed by one Newton step is an effective hybrid for high-precision numerical evaluation.

9.5 Algorithm V: direct inverse in a problem-adapted small parameter

Whenever an exact substitution reduces the problem to an analytic equation $q = \Phi(\tau; \lambda)$ with $\Phi(0; \lambda) = 0$ and $\partial_\tau \Phi(0; \lambda) \neq 0$, one should consider reversing Φ directly. In the present case, $q = v/z$ is exponentially small and (68) is cheaper than generic composition. This method also exposes actual convergence, which a purely formal outer calculation does not automatically prove.

Summary

The five methods are mathematically equivalent but computationally distinct. Lagrange–Bürmann gives closed coefficients; triangular cancellation is the simplest certified block engine; fixed-point iteration is structurally transparent; Newton is fastest at high precision; a problem-adapted inverse such as $q = \Phi_v(\tau)$ can reveal convergence and hidden resummations.

10 Analytic transfer and residual certificates

A formal transseries identity does not, by itself, state an analytic error bound. For monotone real inverse problems, the residual supplies an elementary and often sharp bridge.

Theorem 10.1 (Residual-to-error transfer)**.** *Let F be continuously differentiable and strictly increasing on an interval containing the exact inverse value $G(X)$ and an approximation $H(X)$. If*

$$0 < m \leq F'(y) \leq M \tag{99}$$

throughout the segment between $G(X)$ and $H(X)$, and $R = F(H) - X$, then

$$\frac{|R|}{M} \leq |H - G| \leq \frac{|R|}{m}. \tag{100}$$

Proof. The mean value theorem gives $R = F(H) - F(G) = F'(\xi)(H - G)$ for some intermediate ξ. Apply (99). □

Lean status. The instance used below, $F = f$, $m = 1$, $M = 2$, is formal in `LambertShiftInverse.lean`: the derivative bracket (102) is `Fabius.one_le_deriv_lambertShift` and `Fabius.deriv_lambertShift_le_two` (from `Fabius.lambertShift_hasDerivAt` and the bound $\mathrm{W}' \leq 1$, `Fabius.inv_exp_mul_add_one_le_one`), and the mean-value transfer is `Fabius.abs_sub_lambertShiftInv_le_abs_residual` ($|H - G| \leq |R|$) and `Fabius.abs_residual_le_two_mul_abs_sub_lambertShiftInv` ($|R| \leq 2|H - G|$), for every $z \geq 0$ and every approximation $H \geq 0$. The general (m, M) statement is formal too, for an arbitrary function on an arbitrary convex set, in `MeanValueBracket.lean`: the two brackets are `Fabius.mul_abs_sub_le_abs_sub_of_le_deriv` and `Fabius.abs_sub_le_of_le_deriv`, and the transfer against a right inverse is `Fabius.abs_sub_right_inverse_le_div` ($|H - G| \leq |R|/m$) together with `Fabius.div_le_abs_sub_right_inverse` ($|R|/M \leq |H - G|$), which is (100).

One hypothesis deserves recording, because the theorem as printed leaves it implicit in the words "strictly increasing". The upper bracket needs $0 \le m$, not merely $m \le F' \le M$: with $F' \equiv -10$, $m = -10$ and $M = 0$ the derivative bracket holds while $|F(x) - F(y)| = 10|x - y|$ exceeds $M|x - y| = 0$. What the sign gives is monotonicity, which is what makes $|F(x) - F(y)|$ the signed increment; the printed $0 < m$ supplies it. The lower bracket needs no sign condition.

For $F(x) = x + \mathrm{W}(x)$, (3) gives

$$0 < \mathrm{W}'(x) = \frac{\mathrm{e}^{-\mathrm{W}(x)}}{1 + \mathrm{W}(x)} \le 1, \qquad x \ge 0. \tag{101}$$

Therefore

$$1 \le F'(x) \le 2. \tag{102}$$

For every approximation $H \ge 0$, we obtain the particularly simple certificate

$$\boxed{\frac{1}{2}|H + \mathrm{W}(H) - z| \le |H - g(z)| \le |H + \mathrm{W}(H) - z|.} \tag{103}$$

Thus every symbolic truncation can be checked numerically without first computing the exact inverse.

For an outer truncation

$$H_N(z) = z - v + \sum_{m=1}^{N} \frac{A_m(v)}{z^m}, \tag{104}$$

Theorem 6.1 gives

$$H_N - g = O_N(q^{N+1}). \tag{105}$$

The residual certificate independently verifies the same order and catches coefficient or truncation mistakes in an implementation.

11 Numerical behavior

The following table compares the exact inverse with successive W-anchored truncations. The exact reference value was obtained by solving $ue^u + u = z$ at high precision and using $g = z - u$. The column $M = 0$ is $z - \mathrm{W}(z)$; for $M \ge 1$, the approximation is H_M from (104). All entries after the q column are absolute errors.

z	$q = \mathrm{W}(z)/z$	$M = 0$	$M = 1$	$M = 2$	$M = 4$	$M = 6$
10	1.7455280×10^{-1}	1.1202183×10^{-1}	1.0461658×10^{-3}	2.9896578×10^{-4}	6.8888567×10^{-7}	1.1896934×10^{-8}
10^2	3.3856301×10^{-2}	2.6355095×10^{-2}	2.1861948×10^{-4}	9.4974943×10^{-7}	7.9460859×10^{-10}	5.5483129×10^{-14}
10^4	7.2318460×10^{-4}	6.3550313×10^{-4}	1.7057929×10^{-7}	5.3410695×10^{-11}	4.0832057×10^{-18}	2.9566636×10^{-25}
10^8	1.5668997×10^{-7}	1.4728989×10^{-7}	1.0113357×10^{-14}	8.8743507×10^{-22}	8.3618466×10^{-36}	8.2973506×10^{-50}

Table 1: Absolute error of the W-anchored expansion.

Several features are visible.

- The error of $z - v$ is not a defect in the ordinary Lambert expansion; it is the omitted transseries sector $\tau \sim q$.
- Each additional outer block usually reduces the error by approximately a factor of q, in agreement with (69).
- Even at $z = 10$, where $q \approx 0.175$ is not extremely small, six correction blocks give about eight decimal digits in absolute error.
- At large z, evaluating $v = \mathrm{W}(z)$ once and applying rational corrections is dramatically more efficient than expanding all slow logarithmic subblocks to comparable accuracy.

12 Parameter generalizations

12.1 The family $x + r\,\mathrm{W}(x)$

Let

$$f_r(x) = x + r\,\mathrm{W}(x), \qquad r \geq 0, \tag{106}$$

and let $g_r = f_r^{-1}$. Setting $u = \mathrm{W}(g_r(z))$ gives

$$u\mathrm{e}^u + ru = z, \tag{107}$$

so

$$g_r(z) = z - r\mathcal{W}_r(z). \tag{108}$$

The near-identity formula yields

$$g_r(z) = z + \sum_{n\geq 1} \frac{(-r)^n}{n!} \left(\frac{\mathrm{d}}{\mathrm{d}z}\right)^{n-1} \mathrm{W}(z)^n. \tag{109}$$

In terms of the coefficient functions already computed,

$$\boxed{g_r(z) = z - rv + \sum_{m\geq 1} \frac{r^{m+1} A_m(v)}{z^m}, \qquad v = \mathrm{W}(z).} \tag{110}$$

Equivalently, for $r \neq 0$,

$$\mathcal{W}_r(z) = v - \sum_{m\geq 1} \frac{r^m A_m(v)}{z^m}. \tag{111}$$

The direct equation becomes $rq = \Phi_v(\tau)$, so the same coefficient functions $c_n(v)$ apply after replacing q by rq.

12.2 The scaled family $x + \alpha\,\mathrm{W}(\beta x)$

Let $\alpha, \beta > 0$ and

$$F_{\alpha,\beta}(x) = x + \alpha\,\mathrm{W}(\beta x). \tag{112}$$

If $u = \mathrm{W}(\beta x)$, then $x = u\mathrm{e}^u/\beta$, and the inverse equation is

$$u\mathrm{e}^u + \alpha\beta u = \beta z. \tag{113}$$

Hence

$$F_{\alpha,\beta}^{-1}(z) = z - \alpha\mathcal{W}_{\alpha\beta}(\beta z). \tag{114}$$

Putting $v = \mathrm{W}(\beta z)$, the same universal rational functions give

$$\boxed{F_{\alpha,\beta}^{-1}(z) = z - \alpha v + \sum_{m\geq 1} \frac{\alpha^{m+1} A_m(v)}{z^m}.} \tag{115}$$

The parameter β enters only through the anchor $v = \mathrm{W}(\beta z)$. The cancellation follows from

$$\frac{\mathrm{d}}{\mathrm{d}z}\,\mathrm{W}(\beta z) = \frac{v}{z(1+v)}.$$

12.3 General slowly varying perturbations

The same machinery applies to

$$F(X) = X + A(X), \qquad A(X) = o(X), \tag{116}$$

whenever derivatives of A lower the asymptotic order by one outer power. Examples include finite combinations of iterated logarithms, rational functions of $\log X$, completed power-logarithmic series, and functions living in a differential Hardy field or a suitable logarithmic-exponential transseries field. The essential structural hypotheses are:

1. a complete outer filtration in which A has order zero relative to $t = X^{-1}$;
2. a derivation D that raises outer valuation;
3. locally finite Taylor composition by $X + B$;
4. invertibility of the leading derivative, here the unit $F' = 1 + O(t)$.

Under these conditions, (31), (35), and the residual algorithms remain valid verbatim.

13 Further deductions and structural observations

13.1 An expansion of the generalized Lambert value itself

Since $u = \mathcal{W}_1(z) = z - g(z)$, Theorem 5.1 immediately gives

$$\mathcal{W}_1(z) = v - \sum_{m \geq 1} \frac{A_m(v)}{z^m}. \tag{117}$$

Thus

$$\begin{aligned}\mathcal{W}_1(z) = v &- \frac{v^2}{z(1+v)} - \frac{v^3(v^2-2)}{2z^2(1+v)^3} \\ &- \frac{v^4(2v-1)(v^3-6v-6)}{6z^3(1+v)^5} + O\left(\left(\frac{v}{z}\right)^4\right).\end{aligned} \tag{118}$$

This refines the leading generalized-Lambert asymptotics while keeping the ordinary Lambert function as an exact comparison solution.

13.2 The correction as a perturbation of Wright omega

The equations for u and v may also be written

$$v + \log v = L, \tag{119}$$

$$u + \log u + \log(1 + \mathrm{e}^{-u}) = L. \tag{120}$$

The first equation says $v = \omega(L)$, where ω is the Wright omega function. The last logarithm in (120) is exponentially small in u; it is precisely the source of the missing transseries sector. This provides a second conceptual explanation for why the complete inverse-logarithm expansion of u agrees with that of v, while the exact functions differ.

13.3 Closure under differentiation

The anchored coefficient field $\mathbb{Q}(v)$ is stable under the slow derivation ϑ. Therefore the complete transseries (43) may be differentiated term by term in its domain of convergence. For example,

$$g'(z) = 1 - \frac{v}{z(1+v)} + \frac{1}{z^2}(\vartheta - 1)A_1(v) + \frac{1}{z^3}(\vartheta - 2)A_2(v) + \cdots. \tag{121}$$

This must agree with the exact formula (12) after substituting $u = v - \tau$. The equality supplies a useful independent check on computed coefficient blocks.

13.4 Potential coefficient questions

The rational functions A_m, or equivalently the integer polynomials P_m in (57), invite further study. Among the natural questions are:

- determining sharp bounds and the zero geometry of P_m;
- finding direct combinatorial interpretations of its coefficients;
- deriving a bivariate generating function for the diagonal $R_{m+1,m}$ of (55);
- locating the exact singularities of Φ_v^{-1} and hence the exact convergence radius of (60);
- formalizing the filtered reversion theorem and residual certificates in a proof assistant, with coefficient calculations reflected into exact rational identities.

These questions are not needed for the asymptotic expansion, but they may lead to a useful polynomial theory associated with the r-Lambert function.

14 Conclusions

The inverse of $x + \mathrm{W}(x)$ is simple enough to admit an exact generalized special-function description and rich enough to expose a genuine transseries phenomenon.

First, the substitution $u = \mathrm{W}(x)$ turns the inverse problem into $ue^u + u = z$, so $g(z) = z - \mathcal{W}_1(z)$. Second, anchoring at the ordinary Lambert value $v = \mathrm{W}(z)$ produces the all-orders expansion

$$g(z) = z - v + \sum_{m\geq 1} z^{-m} A_m(v),$$

with the explicit operator formula (44). Third, the exact small-parameter equation

$$\frac{v}{z} = \frac{v}{v - \tau} - \mathrm{e}^{-\tau}$$

shows that this outer series converges for sufficiently large z, gives the remainder scale $(v/z)^{N+1}$, and reveals harmonic-number structure after a second expansion in $1/v$. Finally, the classical inverse-logarithm series is recovered by expanding v, but that rank-one series necessarily misses $\tau \sim (\log z)/z$, which is beyond all its orders.

The general lesson is operational. For a logarithmic near-identity map $F = X + A$, do not guess the inverse coefficient by coefficient in a collection of unrelated logarithmic monomials. Place the slow coefficients in a differential algebra, use the outer variable $t = X^{-1}$, exploit the fact that differentiation raises outer valuation, and apply the locally finite formula

$$F^{-1} = X + \sum_{n\geq 1} \frac{(-1)^n}{n!} D^{n-1} A^n.$$

The block formula (35), triangular residual cancellation, Newton iteration, and residual-to-error transfer then provide a complete chain from formal derivation to certified numerical approximation.

A Coefficient tables

For reference, write

$$A_m(v) = \frac{v^{m+1} P_m(v)}{m!(1 + v)^{2m-1}}. \tag{122}$$

The first six numerator polynomials are

$$\begin{aligned}
P_1(v) &= 1,\\
P_2(v) &= v^2 - 2,\\
P_3(v) &= (2v-1)(v^3 - 6v - 6),\\
P_4(v) &= 6v^6 - 8v^5 - 69v^4 - 40v^3 + 96v^2 + 72v - 24,\\
P_5(v) &= 24v^8 - 58v^7 - 428v^6 - 151v^5 + 1320v^4 + 1380v^3\\
&\quad - 480v^2 - 720v + 120,\\
P_6(v) &= 120v^{10} - 444v^9 - 2920v^8 + 424v^7 + 16577v^6 + 17892v^5\\
&\quad - 13530v^4 - 26400v^3 - 1800v^2 + 7200v - 720.
\end{aligned} \tag{123}$$

They satisfy

$$\deg P_m = 2m-2, \qquad \mathrm{lc}(P_m) = (m-1)!, \qquad P_m(0) = (-1)^{m+1} m!.$$

The first eight slow Lambert polynomials in $\mathrm{W}(z) = L - \ell + \sum_{n\geq 1} P^{\mathrm{L}}_n(\ell) L^{-n}$ are

$$\begin{aligned}
P^{\mathrm{L}}_1(y) &= y,\\
P^{\mathrm{L}}_2(y) &= \frac{y(y-2)}{2},\\
P^{\mathrm{L}}_3(y) &= \frac{y(2y^2 - 9y + 6)}{6},\\
P^{\mathrm{L}}_4(y) &= \frac{y(3y^3 - 22y^2 + 36y - 12)}{12},\\
P^{\mathrm{L}}_5(y) &= \frac{y(12y^4 - 125y^3 + 350y^2 - 300y + 60)}{60},\\
P^{\mathrm{L}}_6(y) &= \frac{y(20y^5 - 274y^4 + 1125y^3 - 1700y^2 + 900y - 120)}{120},\\
P^{\mathrm{L}}_7(y) &= \frac{y}{840}\big(120y^6 - 2058y^5 + 11368y^4 - 25725y^3\\
&\qquad + 24500y^2 - 8820y + 840\big),\\
P^{\mathrm{L}}_8(y) &= \frac{y}{2520}\big(315y^7 - 6534y^6 + 45962y^5 - 142149y^4\\
&\qquad + 205800y^3 - 135240y^2 + 35280y - 2520\big).
\end{aligned} \tag{124}$$

The superscript L is used only in this appendix to distinguish the Lambert asymptotic polynomials from the numerator polynomials P_m in (122).

B Derivation of the harmonic correction

This appendix records the coefficient extraction behind (79). With $E = 1 - q$ and $T_0 = -\log E$, equations (73)–(74) become

$$\begin{aligned}
T_1 &= -\frac{T_0}{E},\\
T_2 &= \frac{T_0 + T_0^2/2}{E^2} - \frac{T_0^2}{E}.
\end{aligned} \tag{125}$$

The standard coefficient identities are

$$[q^n]T_0 = \frac{1}{n},$$
$$[q^n]\frac{T_0}{E} = H_n,$$
$$[q^n]\frac{T_0^2}{E} = H_n^2 - H_n^{(2)}. \tag{126}$$

To handle T_0^2/E^2, differentiate:

$$\frac{\mathrm{d}}{\mathrm{d}q}\left(\frac{T_0^2}{E}\right) = \frac{2T_0 + T_0^2}{E^2}. \tag{127}$$

Combining (125) and (127) gives

$$[q^n]T_2 = H_n + \frac{n-1}{2}\big(H_n^2 - H_n^{(2)}\big), \tag{128}$$

which proves the coefficient of v^{-2} in (79).

C Reproducible symbolic computation

The following short SymPy program generates the exact rational functions $A_m(v)$ directly from the operator formula (44). It uses exact integer and rational arithmetic.

```
import sympy as sp

v = sp.symbols("v")

def theta(expr):
    """The slow derivative v/(1+v) * d/dv."""
    return sp.cancel(v * sp.diff(expr, v) / (1 + v))

def inverse_block(m: int):
    """Return A_m(v) in g=z-W(z)+sum A_m(W(z))/z**m."""
    if m < 1:
        raise ValueError("m must be positive")
    expr = v ** (m + 1)
    for j in range(m):
        expr = sp.cancel(theta(expr) - j * expr)
    return sp.factor((-1) ** (m + 1) * expr / sp.factorial(m + 1))

for m in range(1, 7):
    print(m, inverse_block(m))
```

A numerical check may solve the monotone equation $ue^u + u = z$ and compare against successive blocks:

```
import mpmath as mp

mp.mp.dps = 80

def exact_inverse(z):
    z = mp.mpf(z)
    v = mp.lambertw(z)
    u = mp.findroot(lambda y: y * mp.exp(y) + y - z,
                    v - v / z)
    return z - u
```

```
# Convert each symbolic A_m to an mpmath callable once.
A = {m: sp.lambdify(v, inverse_block(m), "mpmath")
     for m in range(1, 7)}

def asymptotic_inverse(z, order):
    z = mp.mpf(z)
    vv = mp.lambertw(z)
    result = z - vv
    for m in range(1, order + 1):
        result += A[m](vv) / z**m
    return result

for z in [10, 100, 10_000, 10**8]:
    exact = exact_inverse(z)
    for order in [0, 1, 2, 4, 6]:
        approx = (mp.mpf(z) - mp.lambertw(z)
                  if order == 0
                  else asymptotic_inverse(z, order))
        print(z, order, abs(approx - exact))
```

For a production implementation, one should cache polynomial derivatives, truncate after every arithmetic operation, and verify the final result by the residual bound (103).

D A compact formula sheet

For convenience, the principal identities are collected here.

Exact reduction: $u = \mathcal{W}_1(z), \quad u\mathrm{e}^u + u = z, \quad g = z - u.$

Anchor: $v = \mathrm{W}(z), \quad z = v\mathrm{e}^v, \quad q = \mathrm{e}^{-v} = v/z.$

Small displacement: $\tau = v - u, \quad q = \dfrac{v}{v-\tau} - \mathrm{e}^{-\tau}, \quad g = z - v + \tau.$

Outer coefficient: $A_m(v) = \dfrac{(-1)^{m+1}}{(m+1)!}\displaystyle\prod_{j=0}^{m-1}(\vartheta - j)v^{m+1}, \quad \vartheta = \dfrac{v}{1+v}\dfrac{\mathrm{d}}{\mathrm{d}v}.$

Outer expansion: $g = z - v + \displaystyle\sum_{m\geq 1}\frac{A_m(v)}{z^m}.$

Direct inverse coefficient: $c_n(v) = \dfrac{1}{n}[\tau^{n-1}]\left(\dfrac{\tau}{v/(v-\tau) - \mathrm{e}^{-\tau}}\right)^n, \quad A_n = v^n c_n.$

Remainder: $g - z + v - \displaystyle\sum_{m=1}^{N}\frac{A_m(v)}{z^m} = O_N\big((v/z)^{N+1}\big).$

Slow expansion: $g \sim z - L + \ell - \displaystyle\sum_{n\geq 1}\frac{P_n(\ell)}{L^n}.$

General reversion: $\big(X + A(X)\big)^{-1} = X + \displaystyle\sum_{n\geq 1}\frac{(-1)^n}{n!}D^{n-1}A(X)^n.$

Certification: $\dfrac{1}{2}|H + \mathrm{W}(H) - z| \leq |H - g(z)| \leq |H + \mathrm{W}(H) - z|.$

Acknowledgment of source context

This note was prepared for the *series-and-transseries* research-draft area of the ProveIt repository. The surrounding corpus contains several independent treatments of polynomial-logarithmic transseries, composition, and series reversal. The present article is standalone: all notation, proofs, coefficient formulas, and verification procedures needed for the specific inverse problem are included here.

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